

$$\frac{d\vec{x}}{dt} = \vec{v}, \quad (1)$$

$$\frac{d\vec{v}}{dt} = \frac{q_s}{m_s}(\vec{E} + \vec{v} \times \vec{B}) = \frac{\vec{F}}{m_s}, \quad (2)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad (\text{Faraday's law}) \quad (3)$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}, \quad (\text{Ampère's law}) \quad (4)$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \quad (\text{Gauss's law}) \quad (5)$$

$$\vec{\nabla} \cdot \vec{B} = 0, \quad (\text{No magnetic monopole}) \quad (6)$$

1 The Particle-in-Cell Loop

The PIC loop is at the core of the method. The schema is shown in fig. 1. It consists of the following steps:

1. **Initialization.** When starting the simulation, we sample a discrete amount of *macro-particles* of each species we want to simulate from their respective initial distribution functions. We initialize the positions and velocities of these particles. Furthermore, we also assign all useful particle attributes to these, for example charge, mass, particle ID, etc.
2. **Scatter.** This operation is the bridge from the particles in phase space to the grid. We interpolate attributes which we are interested in from the particle positions to the grid. For example, in the electromagnetic case, we would like to have the charge density ρ and the current density $\vec{J}$ on the grid, since those are the quantities we will need in the next stage.
3. **Solve.** Once we have obtained the charge and current densities ρ and $\vec{J}$, we can solve the Maxwell equations (eqs. (3) to (6)) to obtain the electric and magnetic fields of the particle bunch. These $\vec{E}$ and $\vec{B}$ will be on the grid too.
4. **Gather.** We interpolate the electromagnetic fields $\vec{E}$ and $\vec{B}$ back to the particles.
5. **Push.** Now that we have $\vec{E}$ and $\vec{B}$ on the particles positions, we push them to the next time-step using the Newton-Lorentz equations of motion (eqs. (1) and (2)) in combination with a time integration scheme.

We repeat steps 2-5 until we reach the final time of the simulation.



Figure 1: PIC loop